

Literature Status: The Proposed Hyperbolic Dirichlet-Spectrum Annulus Is Too Large

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Status: full negative answer implied by later literature. For the admissible constant weight $h \equiv 1$, the proposed annulus has outer radius $1/\mu > 1$. A later theorem gives $r_{\mathcal{D}}(C_{\varphi}) = 1$, so that annulus cannot be the spectrum.

1. Original question

Chalendar, Gallardo-Gutiérrez, and Partington study the classical Dirichlet space

$$\mathcal{D} = \left\{ f \in H(\mathbb{D}) : |f(0)|^2 + \int_{\mathbb{D}} |f'(z)|^2 dA(z) < \infty \right\}$$

and weighted composition operators $W_{h,\varphi} = hC_{\varphi}$. Their final remark after Theorem 3.3 asks the following question on arXiv PDF page 13 [1].

Source question. *Let φ be a hyperbolic automorphism with attractive and repelling fixed points $a, b \in \partial\mathbb{D}$, conjugate to*

$$\psi(z) = \frac{(1+\mu)z + (1-\mu)}{(1-\mu)z + (1+\mu)}, \quad 0 < \mu < 1.$$

If $h \in \mathcal{M}(\mathcal{D})$ is continuous at a, b and $W_{h,\varphi}$ is invertible, must

$$\sigma(W_{h,\varphi}) = \left\{ z : \min\{|h(a)|, |h(b)|\}\mu \leq |z| \leq \frac{\max\{|h(a)|, |h(b)|\}}{\mu} \right\}?$$

Theorem 3.3 of the source proves only containment in this annulus.

2. Decisive later result

Eklund, Lindström, and Mleczko subsequently sharpened that bound. Their Theorem 5.2 (arXiv PDF page 14) states the following for hyperbolic automorphisms [2].

Later theorem (Eklund–Lindström–Mleczko). *If hC_{φ} is invertible on $\mathcal{D}$ and $h \in A(\mathbb{D})$, then*

$$r_{\mathcal{D}}(hC_{\varphi}) = \max\{|h(a)|, |h(b)|\},$$

and

$$\sigma_{\mathcal{D}}(hC_{\varphi}) \subseteq \{\lambda : \min\{|h(a)|, |h(b)|\} \leq |\lambda| \leq \max\{|h(a)|, |h(b)|\}\}.$$

The disc-algebra assumption is more than enough for the constant multiplier used below.

3. Counterexample to the proposed equality

Set $h \equiv 1$. Then $h \in \mathcal{M}(\mathcal{D}) \cap A(\mathbb{D})$, it is continuous at a, b and bounded away from zero, and C_φ is invertible with inverse $C_{\varphi^{-1}}$. The later theorem gives

$$r_{\mathcal{D}}(C_\varphi) = 1.$$

If the equality proposed in the source question held, however, then

$$\sigma(C_\varphi) = \{z : \mu \leq |z| \leq 1/\mu\},$$

whose spectral radius is $1/\mu > 1$. This contradicts the later theorem. Therefore the answer to the source question is *no*.

This is a complete answer to the literal question, but it does not compute the exact spectrum for general h . The later paper leaves that sharper problem open when $|h(a)| \neq |h(b)|$.

4. Provenance

The implication above is a direct specialization of a separate later theorem, not a new result. The 2016 paper describes its contribution as an improvement of the 2015 hyperbolic estimate and records the resulting sharper spectral-radius formula; the present note identifies that formula as a negative answer to the earlier annulus-equality question.

References

- [1] I. Chalendar, E. A. Gallardo-Gutiérrez, and J. R. Partington, *Weighted composition operators on the Dirichlet space: boundedness and spectral properties*, Math. Ann. **363** (2015), 1265–1279; arXiv:1503.01130.
- [2] T. Eklund, M. Lindström, and P. Mleczko, *Spectral properties of weighted composition operators on the Bloch and Dirichlet spaces*, Studia Math. **232** (2016), 95–112; arXiv:1602.05805, Theorem 5.2.